

Dao T. Bui

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EDUCATION

University of South Carolina, Honors College

Columbia, South Carolina

B.S. Computer Science (Artificial Intelligence Concentration and Mathematics Minor)

May 2028

- **GPA:** 4.0 / 4.0
- **Relevant Coursework:** Unix & Linux, Data Structures and Algorithms, Computer Architecture, Digital Logic Design, Linear Algebra, Probability & Statistics
- **Scholarships:** Palmetto Fellows (\$40,000), Dean's Scholarship (\$6,000), Presidential Scholarship (\$2,000)

EXPERIENCE

Adaptive Real-Time Systems Lab, University of South Carolina

Columbia, South Carolina

Machine Learning Research Assistant

Jun 2025 – Present

- Developed a computer vision pipeline using **YOLOv8** and **OpenCV** for real-time spatter detection in laser powder bed fusion.
- Trained and benchmarked models using **PyTorch** on UofSC's **Hyperion HPC cluster**, submitting and managing jobs via the **Slurm** scheduler to optimize GPU utilization and throughput.
- Preparing a first-author paper for submission to the **SPIE Journal**, with planned deployment of the trained model on a **Raspberry Pi** for edge inference and process monitoring.

Kappa Theta Pi Fraternity Website

Columbia, South Carolina

Front-End Developer

Sep 2025 – Present

- Built the fraternity's official website using a React/Next.js + TypeScript stack, structuring reusable UI components and hooks aligned with scalable frontend architecture.
- Deployed public-facing pages that helped onboard 4 nonprofit partners and established the website as a primary channel for external recognition and donations to Kappa Theta Pi.
- Collaborated in a branch-based Git workflow with pull requests and code reviews, contributing to a codebase with 40+ commits and active feature development.

Brain Cognition Lab, Medical University of South Carolina

Charleston, South Carolina

Research Intern

Jun 2024 – Jul 2024

- Automated analysis of cognitive-test datasets ($N \approx 220$) using **Python**, scripting reusable workflows to identify performance trends and early-detection indicators in Alzheimer's patients.
- Developed Python-based preprocessing pipelines for brain-scan data, applying batch transformations to reduce noise, validate scan quality, and standardize outputs for downstream analysis.
- Presented findings on facial recognition and episodic memory as early Alzheimer's screening tools at the **SC Alzheimer's Disease Research Center Symposium**.

PROJECTS

Computer Vision Sobriety Test | *Python, FastAPI, Mediapipe, OpenAI API*

Sep 2025

- Developed a full-stack AI sobriety testing platform using Next.js, TypeScript, and FastAPI, integrating **MediaPipe** for real-time pose and gaze tracking at ~ 30 FPS.
- Leveraged **AI tools** and the **OpenAI API** to analyze gait, iris velocity, and speech clarity—achieving $>85\%$ accuracy and generating adaptive feedback via a ChatGPT wrapper.

NASA Hacks — SAR Data Analysis App | *Python, FastAPI, AWS, JavaScript*

Sep 2025

- Led a 5-member team to build a full-stack web app analyzing **SAR satellite data** for global oil impact visualization; won **Best Use of Satellite Data** (25+ teams).
- Developed backend data-processing APIs with **FastAPI**.

Full Stack Portfolio Backtester | *FastAPI, Tailwind, Chart.js*

Aug 2025

- Built a FastAPI-powered finance dashboard supporting 50+ assets, 1,200+ trading days, and backtests completed in under 3 seconds.
- Implemented 7+ portfolio strategies with interactive visualizations, metrics (Sharpe Ratio, volatility, 60/40 split), and trend analytics.

TECHNICAL SKILLS

Languages & Frameworks: Python, Java, C++, Bash, PowerShell, HTML/CSS, PyTorch, FastAPI, OpenCV, NumPy
Systems, HPC & Tools: Linux (Ubuntu/RHEL), Slurm, Job Scheduling, Git, SSH, Label Studio